

```
# Initialize Otter
import otter
grader = otter.Notebook("hw6.ipynb")
```

CPSC 330 - Applied Machine Learning

Homework 6: Clustering

Associated lectures: [Lectures 14 and 15](#)

Due date: See [deliverable due dates](#).

Imports

```
import os
from hashlib import sha1

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd

%matplotlib inline
pd.set_option("display.max_colwidth", 0)
```

Points: 4

Exercise 1: Document clustering warm-up

In this homework, we will explore a popular application of clustering called **document clustering**. A large amount of unlabeled text data is available out there (e.g., news, recipes, online Q&A, tweets), and clustering is a commonly used technique to organize this data in a meaningful way.

As a warm up, in this exercise you will cluster sentences from a toy corpus. Later in the homework you will work with a real corpus.

The code below extracts introductory sentences of Wikipedia articles on a set of queries. To run the code successfully, you will need the `wikipedia` package installed in the course environment.

```
conda activate cpsc330
conda install -c conda-forge wikipedia
```

Your tasks:

Run the code below which

- extracts content of Wikipedia articles on a set of queries
- tokenizes the text (i.e., separates sentences) and
- stores the 2nd sentence in each article as a document representing that article

Feel free to experiment with Wikipedia queries of your choice. But stick to the provided list for the final submission so that it's easier for the TAs to grade your submission.

For tokenization we are using the `nltk` package. If you do not have this package in the course environment, you will have to install it.

```
conda activate cpsc330
conda install -c anaconda nltk
```

Even if you have the package installed via the course `conda` environment, you might have to download `nltk` pre-trained models, which can be done with the code below.

```
import nltk

nltk.download("punkt")
nltk.download('punkt_tab')

[nltk_data] Downloading package punkt to
[nltk_data] C:\Users\zhouz\AppData\Roaming\nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to
[nltk_data] C:\Users\zhouz\AppData\Roaming\nltk_data...
[nltk_data] Package punkt_tab is already up-to-date!

True

import wikipedia
import string
from nltk.tokenize import sent_tokenize, word_tokenize

queries = [
    "Artificial Intelligence",
    "Deep learning",
    "Unsupervised learning",
    "Quantum Computer",
    "Environmental protection",
    "Climate Change",
    "Renewable Energy",
    "Biodiversity",
    "French Cuisine",
    "Bread food",
    "Dumpling food",
]
```

```

wiki_dict = {"wiki query": [], "text": [], "n_words": []}
remove_tokens = list(string.punctuation) + ['`', "'", '"', 'br', '"',
'"', '"', '"', 's', "(", ")", "[", "]"]

# Running this code might take some time.
for query in queries:
    try:
        # Attempt to fetch the page content
        page_content = wikipedia.page(query).content
    except wikipedia.exceptions.PageError:
        print(f"Page not found for query: {query}. Skipping...")
        continue
    except wikipedia.exceptions.DisambiguationError as e:
        print(f"Query: {query} led to a disambiguation page. Choosing
the first option: {e.options[0]}")
        page_content = wikipedia.page(e.options[0]).content

    text = sent_tokenize(page_content)[0]
    tokenized = word_tokenize(text)
    text_pp = [token.lower() for token in tokenized if token.lower()
not in remove_tokens]
    wiki_dict["n_words"].append(len(text_pp))
    wiki_dict["text"].append(" ".join(text_pp))
    wiki_dict["wiki query"].append(query)

wiki_df = pd.DataFrame(wiki_dict)
wiki_df

```

```

          wiki query \
0  Artificial Intelligence
1  Deep learning
2  Unsupervised learning
3  Quantum Computer
4  Environmental protection
5  Climate Change
6  Renewable Energy
7  Biodiversity
8  French Cuisine
9  Bread food
10 Dumpling food

```

```

text \
0  artificial intelligence ai is the capability of computational
systems to perform tasks typically associated with human intelligence
such as learning reasoning problem-solving perception and decision-
making
1  in machine learning deep learning dl focuses on utilizing
multilayered neural networks to perform tasks such as classification
regression and representation learning

```

2 in machine learning supervised learning sl is a type of machine learning paradigm where an algorithm learns to map input data to a specific output based on example input-output pairs

3 a quantum computer is a real or theoretical computer that exploits superposed and entangled states

4 environmental protection or environment protection refers to the taking of measures to protecting the natural environment prevent pollution and maintain ecological balance

5 present-day climate change includes both global warming – the ongoing increase in global average temperature – and its wider effects on earth climate system

6 renewable energy also called green energy is energy made from renewable natural resources that are replenished on a human timescale

7 biodiversity is the variability of life on earth

8 french cuisine is the cooking traditions and practices of france

9 bread is a baked food product made from water flour and often yeast

10 manti is a type of dumpling mainly found in turkish cuisine armenian cuisine and central asian cuisine but also in west asia south caucasus and the balkans

	n_words
0	25
1	22
2	30
3	15
4	22
5	24
6	20
7	8
8	10
9	13
10	27

Our toy corpus has six toy documents (`text` column in the dataframe) extracted from Wikipedia queries.

1.1 How many clusters?

rubric={reasoning}

Points:1

Number of clusters: 4

Cluster 1: Artificial Intelligence / Machine Learning Cluster 2: Environment / Climate Cluster 3: Food / Cuisine Cluster 4: Quantum Computing

1.2 K-Means with bag-of-words representation

rubric={accuracy}

In the lecture, we explored how **data representation significantly impacts clustering quality**. We saw that replacing a **flattened image representation** with feature vectors extracted from **pre-trained models** greatly improved clustering results.

What about text data? In previous machine learning courses, we have used the **bag-of-words (BoW) representation** to numerically encode text, where each document is represented with a vector of word frequencies.

In this exercise, you will **cluster documents** using this **simplistic text representation**.

Points:2

```
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.cluster import KMeans

vectorizer = CountVectorizer()
X = vectorizer.fit_transform(wiki_df["text"])

kmeans = KMeans(n_clusters=4, random_state=42)
kmeans_bow_labels = kmeans.fit_predict(X)

wiki_df["bow_kmeans"] = kmeans_bow_labels
wiki_df
```

	wiki query \
0	Artificial Intelligence
1	Deep learning
2	Unsupervised learning
3	Quantum Computer
4	Environmental protection
5	Climate Change
6	Renewable Energy
7	Biodiversity

8 French Cuisine
9 Bread food
10 Dumpling food

text \

0 artificial intelligence ai is the capability of computational systems to perform tasks typically associated with human intelligence such as learning reasoning problem-solving perception and decision-making

1 in machine learning deep learning dl focuses on utilizing multilayered neural networks to perform tasks such as classification regression and representation learning

2 in machine learning supervised learning sl is a type of machine learning paradigm where an algorithm learns to map input data to a specific output based on example input-output pairs

3 a quantum computer is a real or theoretical computer that exploits superposed and entangled states

4 environmental protection or environment protection refers to the taking of measures to protecting the natural environment prevent pollution and maintain ecological balance

5 present-day climate change includes both global warming – the ongoing increase in global average temperature – and its wider effects on earth climate system

6 renewable energy also called green energy is energy made from renewable natural resources that are replenished on a human timescale

7 biodiversity is the variability of life on earth

8 french cuisine is the cooking traditions and practices of france

9 bread is a baked food product made from water flour and often yeast

10 manti is a type of dumpling mainly found in turkish cuisine armenian cuisine and central asian cuisine but also in west asia south caucasus and the balkans

	n_words	bow_kmeans
0	25	1
1	22	2
2	30	2
3	15	1
4	22	0

5	24	1
6	20	3
7	8	1
8	10	1
9	13	1
10	27	1

1.3 K-Means with sentence embedding representation

rubric={accuracy}

In the previous exercise, you may have noticed that clustering results were suboptimal. While the bag-of-words (BoW) representation is useful, it has significant limitations. It does not capture word order or context effectively.

To improve text representation, we can use pre-trained models, similar to those used for image data, to extract richer, context-aware embeddings through transfer learning. In this lab, we will use the **pre-trained model** 'all-MiniLM-L6-v2', available via the [Sentence Transformers](#) package.

This deep learning model generates **dense, fixed-length vector representations** of sentences, capturing **semantic meaning and contextual information**. These representations are particularly effective for **clustering and semantic similarity tasks**. If you're interested in exploring more **pre-trained models**, refer to the [full list here](#). We will explore these representations further in **DSCI 575**.

```
from sentence_transformers import SentenceTransformer

embedder = SentenceTransformer("all-MiniLM-L6-v2")

c:\Users\zhouz\miniforge3\envs\cp330-fresh\lib\site-packages\
transformers\tokenization_utils_base.py:1601: FutureWarning:
`clean_up_tokenization_spaces` was not set. It will be set to `True`
by default. This behavior will be deprecated in transformers v4.45, and
will be then set to `False` by default. For more details check this
issue: https://github.com/huggingface/transformers/issues/31884
  warnings.warn(

emb_sents = embedder.encode(wiki_df["text"].tolist())
emb_sent_df = pd.DataFrame(emb_sents, index=wiki_df.index)
emb_sent_df
```

	0	1	2	3	4	5	
6 \	0	-0.005857	-0.004795	-0.000976	0.011121	0.005294	-0.061630
	0.062072						

1	-0.122905	-0.092450	0.085379	-0.009761	0.001422	0.069146	-
	0.046654						
2	-0.056721	-0.049697	-0.014780	0.022572	0.051773	0.007456	
	0.008632						
3	-0.071932	-0.006760	-0.083421	0.022650	-0.116415	0.034988	-
	0.055792						
4	-0.012700	0.101830	0.066676	-0.007987	0.140040	0.041627	
	0.035495						
5	-0.014444	0.022869	0.145982	0.054865	0.085204	-0.014689	-
	0.102757						
6	-0.044869	0.076882	-0.012749	0.033335	0.067894	-0.002326	
	0.031620						
7	0.027806	0.077117	0.015402	-0.013544	0.043216	0.048124	
	0.027763						
8	0.026616	-0.019929	-0.008049	-0.025907	-0.055372	0.011235	-
	0.024538						
9	-0.014786	0.055887	-0.020880	0.005313	-0.025037	0.040479	
	0.035260						
10	-0.017367	0.046664	-0.089176	0.010790	-0.084132	-0.039986	
	0.033869						
	7	8	9	...	374	375	376
377	\						
0	0.033673	-0.035251	0.074072	...	0.088312	-0.006600	0.036511 -
	0.020797						
1	-0.069960	-0.021858	-0.052803	...	0.065826	0.031800	-0.021052 -
	0.113772						
2	-0.074847	-0.065573	-0.027284	...	0.014761	-0.022392	0.021360 -
	0.107016						
3	-0.045055	-0.017993	0.020915	...	-0.076031	-0.016973	-0.054588 -
	0.020448						
4	-0.050723	-0.037633	0.055886	...	-0.042105	-0.038883	0.018268
	0.009920						
5	0.016853	-0.023799	0.021148	...	0.011778	0.016817	-0.020041
	0.006375						
6	-0.021576	0.046619	0.005089	...	-0.021575	-0.017596	-0.069995
	0.004461						
7	-0.022999	0.049303	0.009686	...	0.014946	-0.020896	0.022283 -
	0.012311						
8	-0.065724	0.058374	-0.062895	...	-0.059513	0.077446	-0.059827 -
	0.021454						
9	0.024938	0.017546	0.072433	...	-0.082098	0.037856	-0.020007 -
	0.087252						
10	-0.100715	-0.010708	-0.068042	...	0.020180	0.021404	0.010321 -
	0.017518						
	378	379	380	381	382	383	
0	-0.002082	0.058263	0.061880	0.097383	0.058155	-0.052016	
1	-0.013588	0.056793	-0.013167	0.009044	-0.011912	-0.011387	

2	0.020530	0.045678	0.052755	0.005430	0.022018	-0.010994
3	-0.016833	0.120573	0.027863	0.076044	0.060111	0.057609
4	0.028208	0.052759	-0.009605	-0.004518	0.060853	0.057709
5	0.055428	0.076868	-0.047152	-0.114797	-0.056382	0.022349
6	-0.040409	-0.043067	0.044042	-0.050977	0.056506	0.048708
7	0.014079	0.028085	0.025272	-0.015233	-0.014879	-0.064280
8	0.088102	-0.127180	-0.031399	0.120572	0.072373	-0.058336
9	0.058163	-0.044187	-0.030253	-0.080385	0.075614	-0.070403
10	0.079356	-0.034642	-0.022086	-0.018417	-0.012915	0.004780

[11 rows x 384 columns]

Points:2

```
from sklearn.cluster import KMeans
```

```
kmeans = KMeans(n_clusters=4, random_state=42)
```

```
kmeans_emb_labels = kmeans.fit_predict(emb_sents)
```

```
wiki_df["emb_kmeans"] = kmeans_emb_labels
wiki_df
```

	wiki query \
0	Artificial Intelligence
1	Deep learning
2	Unsupervised learning
3	Quantum Computer
4	Environmental protection
5	Climate Change
6	Renewable Energy
7	Biodiversity
8	French Cuisine
9	Bread food
10	Dumpling food

```
text \
0  artificial intelligence ai is the capability of computational
systems to perform tasks typically associated with human intelligence
such as learning reasoning problem-solving perception and decision-
making
1  in machine learning deep learning dl focuses on utilizing
multilayered neural networks to perform tasks such as classification
regression and representation learning
2  in machine learning supervised learning sl is a type of machine
learning paradigm where an algorithm learns to map input data to a
specific output based on example input-output pairs
```

3 a quantum computer is a real or theoretical computer that exploits superposed and entangled states

4 environmental protection or environment protection refers to the taking of measures to protecting the natural environment prevent pollution and maintain ecological balance

5 present-day climate change includes both global warming – the ongoing increase in global average temperature – and its wider effects on earth climate system

6 renewable energy also called green energy is energy made from renewable natural resources that are replenished on a human timescale

7 biodiversity is the variability of life on earth

8 french cuisine is the cooking traditions and practices of france

9 bread is a baked food product made from water flour and often yeast

10 manti is a type of dumpling mainly found in turkish cuisine armenian cuisine and central asian cuisine but also in west asia south caucasus and the balkans

	n_words	bow_kmeans	emb_kmeans
0	25	1	2
1	22	2	2
2	30	2	2
3	15	1	2
4	22	0	0
5	24	1	0
6	20	3	0
7	8	1	0
8	10	1	1
9	13	1	1
10	27	1	3

1.4 DBSCAN with sentence embedding representation and cosine distance

rubric={accuracy}

Now, let's apply **DBSCAN** to our toy dataset.

Unlike K-Means, which relies on Euclidean distance due to its use of centroids, DBSCAN allows us to experiment with different distance metrics. For text data, **cosine similarity** (or its complement, **cosine distance**) is often more effective than Euclidean distance.

The **cosine distance** between two vectors u and v is defined as:

$$distance_{cosine}(u,v) = 1 - \left(\frac{u \cdot v}{\|u\|_2 \|v\|_2} \right)$$

```
from sklearn.metrics.pairwise import cosine_distances
```

Points:4

```
dist_matrix = cosine_distances(emb_sents)

dist_vals = dist_matrix[np.triu_indices_from(dist_matrix, k=1)]

print("min:", dist_vals.min())
print("max:", dist_vals.max())
print("mean:", dist_vals.mean())

min: 0.5178395
max: 1.0320184
mean: 0.89426917

from sklearn.cluster import DBSCAN

for eps in [0.1, 0.2, 0.3, 0.4, 0.5, 0.6]:
    db = DBSCAN(eps=eps, min_samples=2, metric="cosine")
    labels = db.fit_predict(emb_sents)

    n_clusters = len(set(labels)) - (1 if -1 in labels else 0)
    unique, counts = np.unique(labels, return_counts=True)
    print("eps =", eps, "| clusters =", n_clusters, "| sizes =",
dict(zip(unique, counts)))

eps = 0.1 | clusters = 0 | sizes = {np.int64(-1): np.int64(11)}
eps = 0.2 | clusters = 0 | sizes = {np.int64(-1): np.int64(11)}
eps = 0.3 | clusters = 0 | sizes = {np.int64(-1): np.int64(11)}
eps = 0.4 | clusters = 0 | sizes = {np.int64(-1): np.int64(11)}
eps = 0.5 | clusters = 0 | sizes = {np.int64(-1): np.int64(11)}
eps = 0.6 | clusters = 1 | sizes = {np.int64(-1): np.int64(8),
np.int64(0): np.int64(3)}

from sklearn.cluster import DBSCAN

dbscan = DBSCAN(eps=0.4, min_samples=2, metric="cosine")
```

```
dbscan_emb_labels = dbscan.fit_predict(emb_sents)
```

```
wiki_df["emb_dbscan"] = dbscan_emb_labels  
wiki_df
```

```
          wiki query \  
0  Artificial Intelligence  
1  Deep learning  
2  Unsupervised learning  
3  Quantum Computer  
4  Environmental protection  
5  Climate Change  
6  Renewable Energy  
7  Biodiversity  
8  French Cuisine  
9  Bread food  
10 Dumpling food
```

```
text \  
0  artificial intelligence ai is the capability of computational  
systems to perform tasks typically associated with human intelligence  
such as learning reasoning problem-solving perception and decision-  
making  
1  in machine learning deep learning dl focuses on utilizing  
multilayered neural networks to perform tasks such as classification  
regression and representation learning  
2  in machine learning supervised learning sl is a type of machine  
learning paradigm where an algorithm learns to map input data to a  
specific output based on example input-output pairs  
3  a quantum computer is a real or theoretical computer that exploits  
superposed and entangled states  
4  environmental protection or environment protection refers to the  
taking of measures to protecting the natural environment prevent  
pollution and maintain ecological balance  
5  present-day climate change includes both global warming – the  
ongoing increase in global average temperature – and its wider effects  
on earth climate system  
6  renewable energy also called green energy is energy made from  
renewable natural resources that are replenished on a human timescale  
7  biodiversity is the variability of life on earth  
8  french cuisine is the cooking traditions and practices of france  
9  bread is a baked food product made from water flour and often
```

yeast

10 manti is a type of dumpling mainly found in turkish cuisine armenian cuisine and central asian cuisine but also in west asia south caucasus and the balkans

	n_words	bow_kmeans	emb_kmeans	emb_dbscan
0	25	1	2	-1
1	22	2	2	-1
2	30	2	2	-1
3	15	1	2	-1
4	22	0	0	-1
5	24	1	0	-1
6	20	3	0	-1
7	8	1	0	-1
8	10	1	1	-1
9	13	1	1	-1
10	27	1	3	-1

1.5 Hierarchical clustering with sentence embedding representation

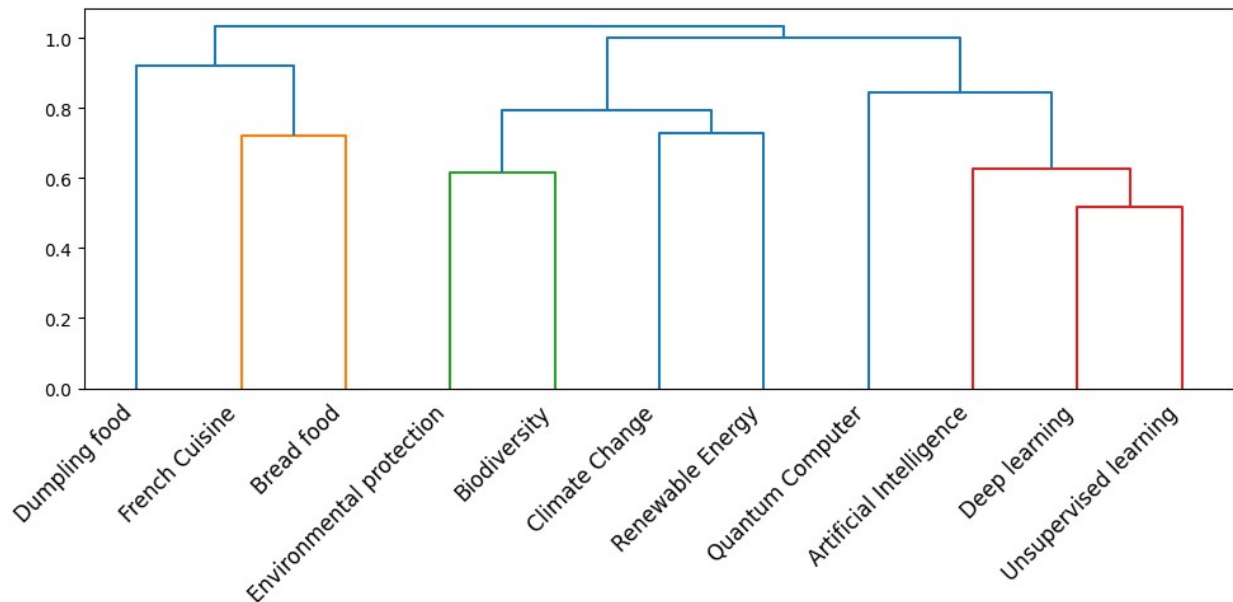
rubric={accuracy}

Points:3

```
from scipy.cluster.hierarchy import linkage, dendrogram, fcluster
import matplotlib.pyplot as plt

Z = linkage(emb_sents, method="complete", metric="cosine")

plt.figure(figsize=(10, 5))
dendrogram(Z, labels=wiki_df["wiki_query"].tolist())
plt.xticks(rotation=45, ha="right")
plt.tight_layout()
plt.show()
```



```
hier_emb_labels = fcluster(Z, t=4, criterion="maxclust")
```

```
wiki_df["emb_hierarchical"] = hier_emb_labels
wiki_df
```

```

      wiki query \
0  Artificial Intelligence
1  Deep learning
2  Unsupervised learning
3  Quantum Computer
4  Environmental protection
5  Climate Change
6  Renewable Energy
7  Biodiversity
8  French Cuisine
9  Bread food
10 Dumpling food

```

```
text \
```

```
0  artificial intelligence ai is the capability of computational
systems to perform tasks typically associated with human intelligence
such as learning reasoning problem-solving perception and decision-
making
```

```
1  in machine learning deep learning dl focuses on utilizing
multilayered neural networks to perform tasks such as classification
regression and representation learning
```

```
2  in machine learning supervised learning sl is a type of machine
learning paradigm where an algorithm learns to map input data to a
specific output based on example input-output pairs
```

3 a quantum computer is a real or theoretical computer that exploits superposed and entangled states

4 environmental protection or environment protection refers to the taking of measures to protecting the natural environment prevent pollution and maintain ecological balance

5 present-day climate change includes both global warming – the ongoing increase in global average temperature – and its wider effects on earth climate system

6 renewable energy also called green energy is energy made from renewable natural resources that are replenished on a human timescale

7 biodiversity is the variability of life on earth

8 french cuisine is the cooking traditions and practices of france

9 bread is a baked food product made from water flour and often yeast

10 manti is a type of dumpling mainly found in turkish cuisine armenian cuisine and central asian cuisine but also in west asia south caucasus and the balkans

	n_words	bow_kmeans	emb_kmeans	emb_dbscan	emb_hierarchical
0	25	1	2	-1	4
1	22	2	2	-1	4
2	30	2	2	-1	4
3	15	1	2	-1	4
4	22	0	0	-1	3
5	24	1	0	-1	3
6	20	3	0	-1	3
7	8	1	0	-1	3
8	10	1	1	-1	1
9	13	1	1	-1	1
10	27	1	3	-1	2

1.6 Discussion

rubric={reasoning}

Points: 4

The clustering results change depending on the representation. BoW only counts words so it does not capture semantic meaning very well. Sentence embeddings work better because they capture similarity between texts.

With embeddings, KMeans and hierarchical clustering group similar topics like AI, environment and food together, which makes sense. DBSCAN was harder to tune since the eps value affects results a lot and many points become noise.

1.7 Visualizing clusters

```
rubric={points:4}
```

One approach to working with unlabeled data is visualization. That said, our data is high-dimensional, making it challenging to visualize. Take sentence embedding representation as an example: each instance is depicted in 768 dimensions. To visualize such high-dimensional data, we can employ dimensionality reduction techniques to extract the most significant 2 or 3 components, and then visualize this low-dimensional data.

Given data as a `numpy` array and corresponding cluster assignments, the `plot_umap_clusters` function below transforms the data by applying dimensionality reduction technique called [UMAP](#) to it and plots the transformed data with different colours for different clusters.

Note: At this point we are using this function only for visualization and you are not expected to understand the UMAP part.

You'll have to install the `umap-learn` package in the course conda environment either with `conda` or `pip`, as described in the [documentation](#).

```
> conda activate cpssc330
> conda install -c conda-forge umap-learn
```

or

```
> conda activate cpssc330
> pip install umap-learn
```

If you get an error with the import below try

```
pip install --upgrade numba umap-learn

import umap

def plot_umap_clusters(
    data,
    cluster_labels,
    raw_texts=None,
    show_labels=False,
    point_size=50,
    n_neighbors=15,
    title="UMAP Visualization",
    ignore_noise=False,
```



```

):
    """
    Perform dimensionality reduction using UMAP and visualize the
    resulting clusters.

    Parameters
    -----
    data : np.ndarray
        Feature matrix to be reduced.
    cluster_labels : array-like
        Cluster label for each data point.
    raw_texts : list of str, optional
        Original texts corresponding to the data points (used when
    show_labels=True).
    show_labels : bool, default=False
        Whether to annotate points with text labels.
    point_size : int, default=50
        Size of the scatterplot points.
    n_neighbors : int, default=15
        Number of neighbors used in UMAP.
    title : str, default="UMAP Visualization"
        Title of the plot.
    ignore_noise : bool, default=False
        Whether to exclude points labeled as noise (-1).

    Returns
    -----
    None
    """
    reducer = umap.UMAP(n_neighbors=n_neighbors, random_state=42)
    embedding = reducer.fit_transform(data)

    df = pd.DataFrame(embedding, columns=["dim1", "dim2"])
    df["cluster"] = cluster_labels

    if ignore_noise:
        df = df[df["cluster"] != -1]

    fig, ax = plt.subplots(figsize=(6, 5))
    ax.set_title(title)

    scatter = ax.scatter(
        df["dim1"], df["dim2"],
        c=df["cluster"], cmap="tab20b", s=point_size
    )

    ax.legend(*scatter.legend_elements(), title="Clusters",
    loc="best")

    if show_labels and raw_texts is not None:

```

```

        for (x, y, text) in zip(df["dim1"], df["dim2"], raw_texts):
            ax.annotate(" ".join(text.split()[:10]), (x, y),
                fontsize=8)

plt.show()

```

Points: 4

```

plot_umap_clusters(
    X.toarray(),
    kmeans_bow_labels,
    raw_texts=wiki_df["text"].tolist(),
    title="K-Means with BoW"
)

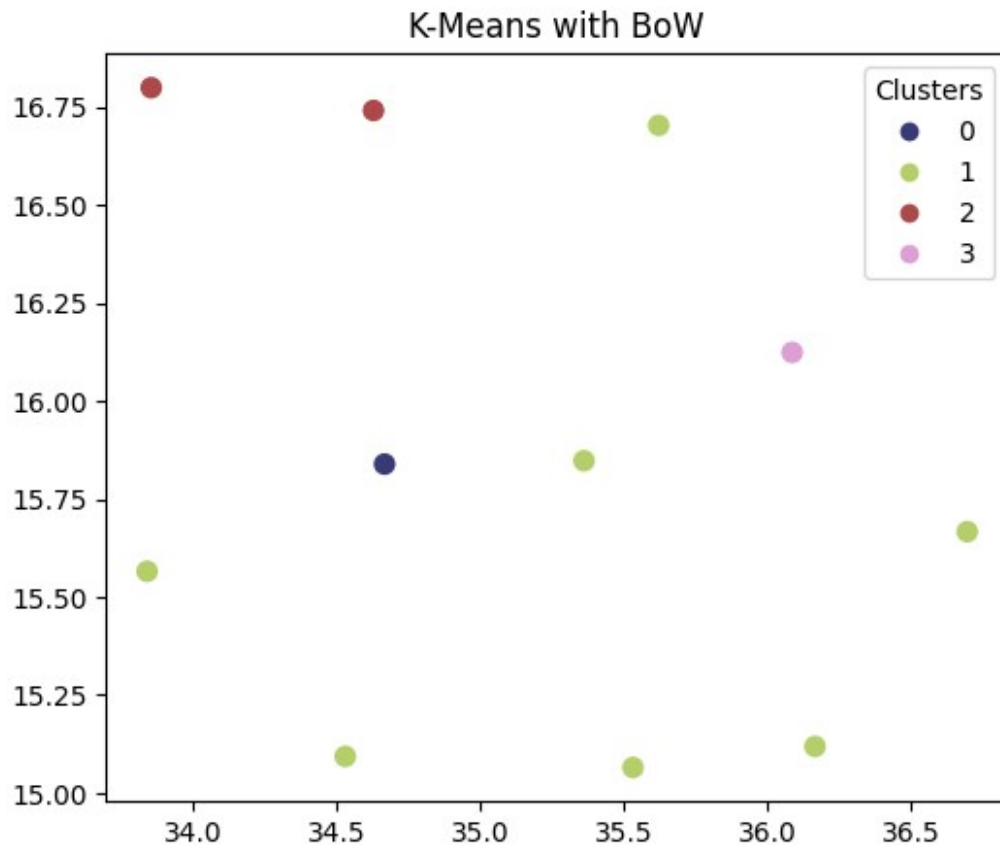
plot_umap_clusters(
    emb_sents,
    kmeans_emb_labels,
    raw_texts=wiki_df["text"].tolist(),
    title="K-Means with Sentence Embeddings"
)

plot_umap_clusters(
    emb_sents,
    dbscan_emb_labels,
    raw_texts=wiki_df["text"].tolist(),
    title="DBSCAN with Sentence Embeddings",
    ignore_noise=False
)

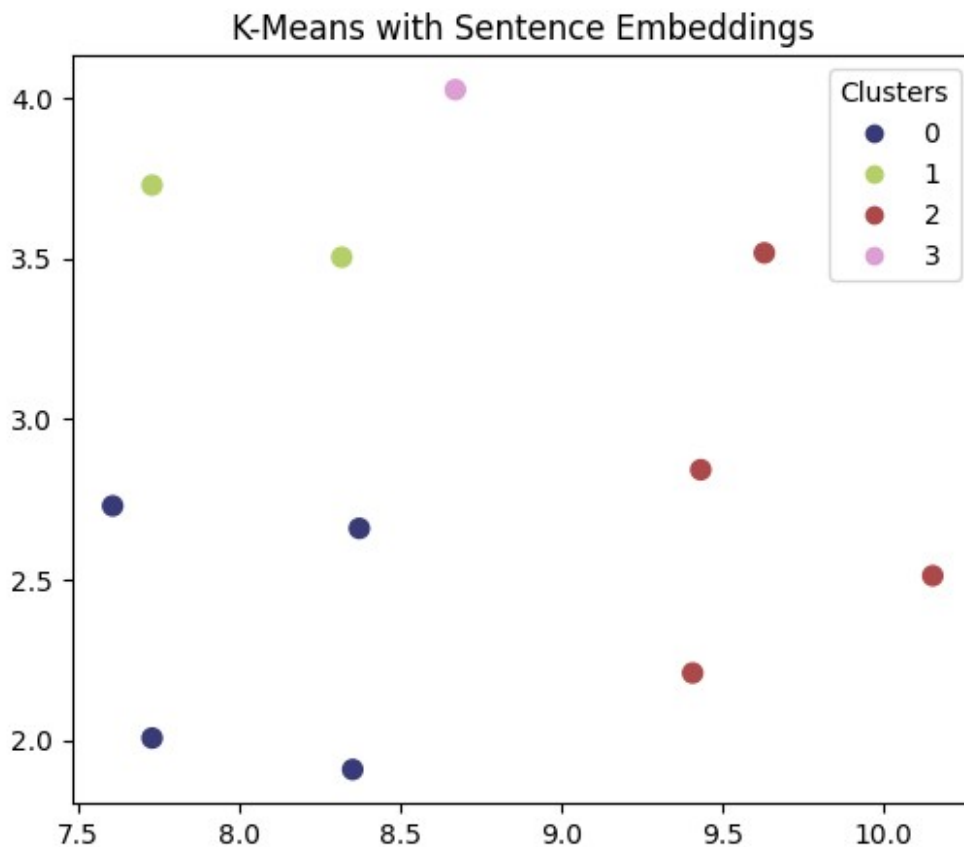
plot_umap_clusters(
    emb_sents,
    hier_emb_labels,
    raw_texts=wiki_df["text"].tolist(),
    title="Hierarchical Clustering with Sentence Embeddings"
)

c:\Users\zhouz\miniforge3\envs\cpssc330-fresh\lib\site-packages\umap\
umap_.py:1952: UserWarning: n_jobs value 1 overridden to 1 by setting
random_state. Use no seed for parallelism.
    warn(
c:\Users\zhouz\miniforge3\envs\cpssc330-fresh\lib\site-packages\umap\
umap_.py:2462: UserWarning: n_neighbors is larger than the dataset
size; truncating to X.shape[0] - 1
    warn(

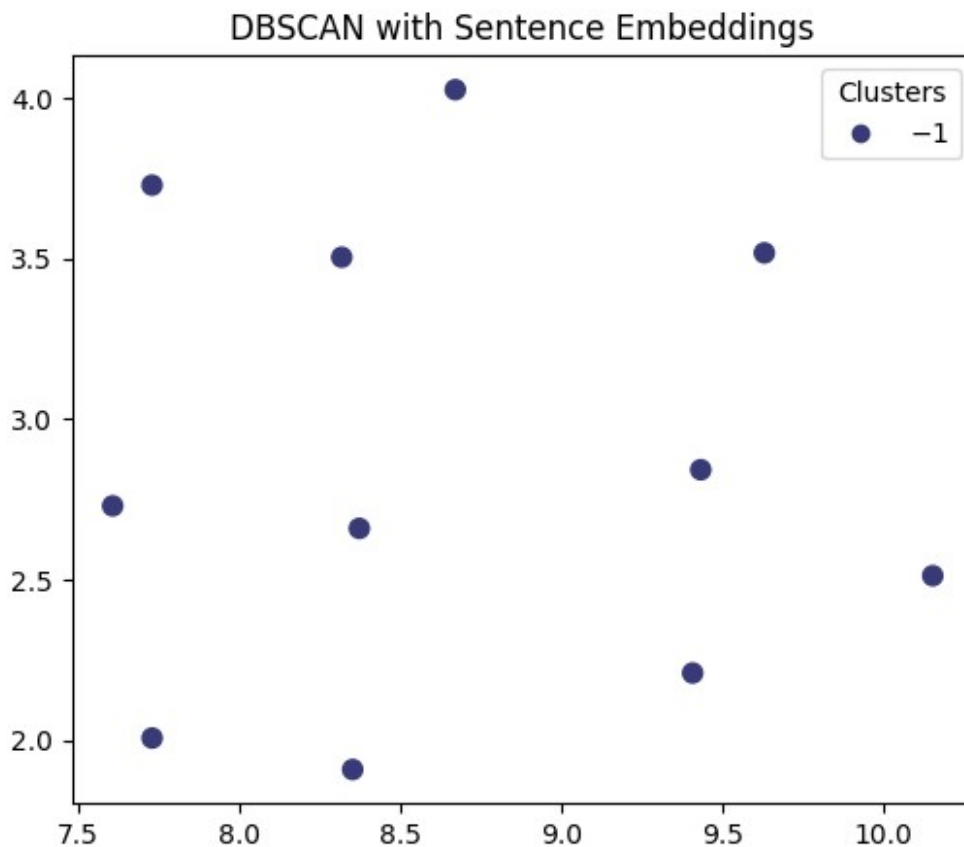
```



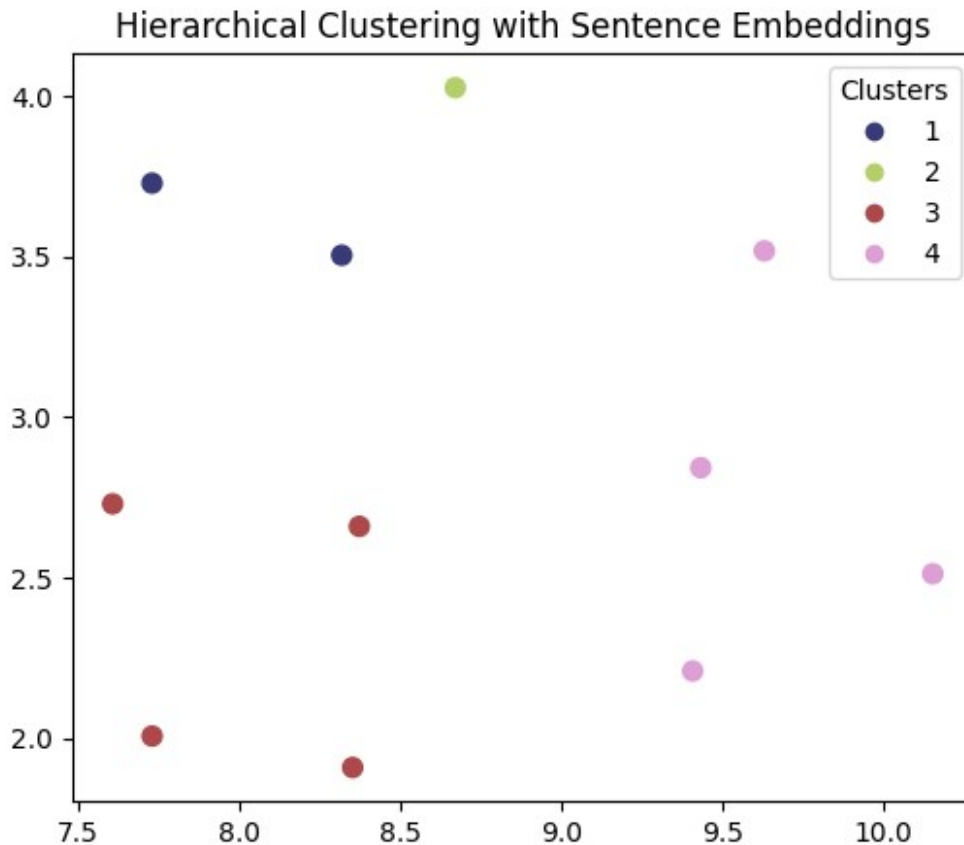
```
c:\Users\zhouz\miniforge3\envs\cpssc330-fresh\lib\site-packages\umap\
umap_.py:1952: UserWarning: n_jobs value 1 overridden to 1 by setting
random_state. Use no seed for parallelism.
  warn(
c:\Users\zhouz\miniforge3\envs\cpssc330-fresh\lib\site-packages\umap\
umap_.py:2462: UserWarning: n_neighbors is larger than the dataset
size; truncating to X.shape[0] - 1
  warn(
```



```
c:\Users\zhouz\miniforge3\envs\cpssc330-fresh\lib\site-packages\umap\
umap_.py:1952: UserWarning: n_jobs value 1 overridden to 1 by setting
random_state. Use no seed for parallelism.
  warn(
c:\Users\zhouz\miniforge3\envs\cpssc330-fresh\lib\site-packages\umap\
umap_.py:2462: UserWarning: n_neighbors is larger than the dataset
size; truncating to X.shape[0] - 1
  warn(
```



```
c:\Users\zhouz\miniforge3\envs\cpssc330-fresh\lib\site-packages\umap\
umap_.py:1952: UserWarning: n_jobs value 1 overridden to 1 by setting
random_state. Use no seed for parallelism.
  warn(
c:\Users\zhouz\miniforge3\envs\cpssc330-fresh\lib\site-packages\umap\
umap_.py:2462: UserWarning: n_neighbors is larger than the dataset
size; truncating to X.shape[0] - 1
  warn(
```



Exercise 2: [Food.com](#) recipes

Now that we have applied document clustering on a toy corpus, let's move to a more realistic corpus.

In the lecture, we worked on an activity of manually clustering food items and discussed challenges associated with it. We also applied different clustering algorithms to cluster food images. We'll continue this theme of clustering food items in this lab. But instead of images we will cluster textual description of food items, i.e., recipe names.

In this assignment, we will work with a sample of [Kaggle's Food.com recipes corpus](#). This corpus contains 180K+ recipes and 700K+ recipe reviews. In this lab, we'll only focus on recipes and **not** on reviews. The recipes are present in `RAW_recipes.csv`. Our goal is to find categories or groupings of recipes from this corpus based on their names.

Your tasks:

- Download `RAW_recipes.csv` and put it under the `data` directory in the homework folder.
- Run the code below. The dataset is quite large, and in this assignment, for speed, you will work with a sample of the dataset. The function `get_recipes_sample` below carries

out some preliminary preprocessing and returns a sample of the recipes with most frequent tags.

Note: Depending upon the capacity of your computer, feel free to increase or decrease the size of this sample by changing the value for `n_tags`. If you decide to go with a different value of `n_tags`, state it clearly in Exercise 2.1 so that the grader knows about it.

```
orig_recipes_df = pd.read_csv("data/RAW_recipes.csv")
orig_recipes_df.shape

(231637, 12)

def get_recipes_sample(orig_recipes_df, n_tags=300, min_len=5):
    orig_recipes_df = orig_recipes_df.dropna() # Remove rows with
    NaNs.
    orig_recipes_df = orig_recipes_df.drop_duplicates(
        "name"
    ) # Remove rows with duplicate names.
    # Remove rows where recipe names are too short (< 5 characters).
    orig_recipes_df =
orig_recipes_df[orig_recipes_df["name"].apply(len) >= min_len]
    # Only consider the rows where tags are one of the most frequent n
    tags.
    first_n = orig_recipes_df["tags"].value_counts()
    [0:n_tags].index.tolist()
    recipes_df =
orig_recipes_df[orig_recipes_df["tags"].isin(first_n)]
    return recipes_df

recipes_df = get_recipes_sample(orig_recipes_df)
recipes_df.shape

(9100, 12)

recipes_df["name"]

42      i yam what i yam  muffins
101      to your health  muffins
129      250 00 chocolate chip cookies
138      lplermagronen
163      california roll  salad
...
231430    zucchini wheat germ cookies
231514    zucchini blueberry bread
231547    zucchini salsa burgers
231596    zuppa toscana
231629    zydeco salad
Name: name, Length: 9100, dtype: object
```

In the rest of the homework, we will use `recipes_df` above, which is a subset of the original dataset.

Points:2

```
shortest_recipe = None
longest_recipe = None

name_lengths = recipes_df["name"].apply(len)

shortest_recipe = recipes_df.loc[name_lengths.idxmin(), "name"]
longest_recipe = recipes_df.loc[name_lengths.idxmax(), "name"]

print("Shortest recipe:", shortest_recipe)
print("Longest recipe:", longest_recipe)

Shortest recipe: bread
Longest recipe: baked tomatoes with a parmesan cheese crust and
balsamic drizzle
```

Points:2

```
from wordcloud import WordCloud
import matplotlib.pyplot as plt

text = " ".join(recipes_df["name"])

wordcloud = WordCloud(width=800, height=400,
background_color="white").generate(text)

plt.figure(figsize=(10,5))
plt.imshow(wordcloud, interpolation="bilinear")
plt.axis("off")
plt.show()
```


231514	-0.052718	0.008980	-0.046014	0.030194	0.005201	0.009964	-
0.006760							
231547	-0.080801	0.004295	-0.044325	0.038308	-0.030125	-0.063566	
0.004788							
231596	-0.060801	0.111672	-0.108755	0.052323	-0.099851	-0.027532	
0.084190							
231629	-0.085059	0.009065	-0.088442	-0.008907	0.002234	-0.055801	
0.039431							

	7	8	9	...	374	375	
376 \							
42	-0.055191	0.020301	0.019828	...	0.063293	-0.067171	
0.087500							
101	0.000466	0.033078	-0.013923	...	-0.012926	-0.015949	
0.031315							
129	0.022929	-0.037398	0.011857	...	0.007971	-0.019165	
0.004935							
138	0.124966	-0.011697	-0.050179	...	-0.042345	-0.005794	-
0.031800							
163	-0.036994	0.023526	-0.046870	...	-0.018432	0.051918	
0.036101							
...	...	...	...	...	...	...	..

231430	0.010435	-0.069214	0.010464	...	0.050988	-0.064541
0.090829						
231514	0.030238	-0.031474	0.024632	...	0.075467	0.000967
0.085033						
231547	0.004822	0.015525	-0.040094	...	0.066642	0.016605
0.096211						
231596	-0.004861	0.002891	0.013943	...	0.038082	-0.014214
0.048392						
231629	-0.037784	0.005315	-0.078361	...	0.063838	0.035654
0.031519						

	377	378	379	380	381	382
383						
42	-0.061550	0.039297	-0.050147	0.027708	0.056843	0.056152
0.122506						
101	-0.059074	0.014143	-0.047270	0.007844	0.035501	0.076061
0.078119						
129	0.009005	0.000919	-0.040078	0.008650	-0.075781	-0.083477
0.123240						
138	0.120664	-0.057335	-0.077068	0.001653	-0.048223	0.116455
0.021789						
163	-0.035312	0.005817	0.101802	-0.063171	-0.007917	0.089744
0.006997						
...	...	...	...	...	...	...
...						
231430	-0.004570	0.079109	0.019663	-0.058483	-0.048723	0.019152
-						

```

0.012266
231514 -0.006520  0.031094  0.072901 -0.094975 -0.052466 -0.003300 -
0.006991
231547 -0.023969  0.045752  0.017091 -0.062939 -0.016950  0.012060
0.039776
231596  0.050377  0.015281  0.106766  0.032009  0.020113  0.004977 -
0.005828
231629 -0.006411  0.007804  0.016261 -0.006218  0.016956  0.019806 -
0.042820

[9100 rows x 384 columns]

```

Exercise 3: Clustering recipe names

In this exercise you'll cluster recipe names with some of the clustering algorithms we have seen in class. This will also involve making some attempts to pick reasonable hyperparameter values for each clustering method based on the quality of the resulting clusters. For example, for KMeans, you need to specify the number of clusters in advance, which is often challenging on real-world datasets. For DBSCAN, you need to pick appropriate `eps` and `min_samples`. For hierarchical clustering, you need to pick a suitable linkage criterion, distance metric, and prune the tree so that it's possible to visualize and interpret it.

Here are some methods which may help you with picking reasonable values for the hyperparameters.

- Visualize the Elbow plot (KMeans).
- Visualize Silhouette plots.
- Visualize resulting clusters using `plot_umap_clusters` function from Exercise 1.
- Sample some recipes from each cluster, manually inspect whether there are coherent semantic themes. (For this, you may use the function `print_clusters` given below.)

You may use the `yellowbrick` package for visualizing the Elbow plot and the Silhouette plots. You can install it with

```
conda install -c districtdatalabs yellowbrick
```

Note that the process of picking reasonable hyperparameter values will be exploratory, iterative, and will involve manual inspection and judgment, as there is no ground truth to verify how well the model is doing. In your solutions, please do not include everything you try. Only present the results of the most informative trials. Add a narrative to your answer so that it's easy for the grader to follow your choices and reasoning.

```

def print_clusters(recipes_df, cluster_labels, n_recipes=10,
replace=False, random_state=None):
    """
    Given recipes_df containing recipe names and cluster assignment
    (labels),
    sample and print n_recipes recipes per cluster.

```

Parameters

recipe_df : pandas dataframe

recipes dataframe containing recipe names in the "name" column

cluster_labels : ndarray or a list

cluster labels for each row in recipes_df

n_recipes : int

number of examples to sample from each cluster

replace: bool

replace flag to pass to the sampling of recipe names

Returns

None

"""

```
grouped = (
    pd.DataFrame(
        {
            "name": recipes_df["name"],
            "cluster_label": cluster_labels,
        }
    )
    .sort_values("cluster_label")
    .groupby("cluster_label")
)

for name, group in grouped:
    print(f"Cluster {name}")
    print(("-----").format(""))
    print("\n".join(group.sample(n_recipes,
random_state=random_state, replace=replace)['name'].tolist()))
    print("\n\n")
```

Points:6

Type your answer here, replacing this text.

```
from sklearn.cluster import KMeans

kmeans = KMeans(n_clusters=12, random_state=42)
kmeans_labels = kmeans.fit_predict(embeddings)

print_clusters(recipes_df, kmeans_labels, n_recipes=10,
random_state=42)
```

Cluster 0

eliz s mushroom risotto
greet the sun breakfast pizzas
stuffed shells bolognese
creamy herb mashed potatoes
mast o musir persian yogurt dip with shallots
red potatoes with a flare
italian spinach pasta bake
gorgonzola focaccia
red pepper feta and garlic hummus
roast potatoes with lemon and coriander

Cluster 1

butter chocolate bliss martini
peanut butter finger bars
kellogg s frosted flakes good times bars
chocolate mint brownies
goldy bear s honey gingersnaps
oriental rice pudding with rhubarb
grilled beer braised bratwursts with cranberry pickle chutney
jam streusel bars
saskatoon berry jam
lemon and blueberry yogurt parfait low fat

Cluster 2

berry cherry banana shortcake
chillin strawberry pie
drumstick cheesecake
lemon whoopie pies
white chocolate rhubarb swirled cheesecake
root beer pancakes
amaretto cheesecake bars
pumpkin hand pies
overnight pumpkin pecan waffles
chocolate fruitcake non alcoholic

Cluster 3

calamondin cake
charlene s 7 up pound cake
corn bread cake
lemon yoghurt cake

coconut mocha cake
east 62nd street lemon cake
butterscotch coffee cake
gelatin poke cake
chocolate marshmallow molten cakes
caramel cake with caramel cream cheese frosting

Cluster 4

linzie s oatmeal raisin delights
oatmeal cookies with white chocolate peanuts and caramel
kossmans oatmeal cookies
whole grain steel cut oatmeal pancakes
oatmeal applesauce muffins
easy oatmeal favorites
granola oats fruits nuts
magnolia cafe oatmeal muffins
chocolate chip oatmeal bars
chocolate chip cookies with oats and toffee bits

Cluster 5

creamy mushroom roasted onion soup
spicy tomato beef soup
machine shed restaurant baked potato soup
potato leek and roasted garlic soup
rich beefy vegetable soup
fennel avgolemono soup
white house broccoli soup
nacho beef soup low carb
spicy moroccan lentil soup
special mexican chicken soup

Cluster 6

honey and garlic dressing 2
baby green salad
tapenade caesar salad with prosciutto
enlightened caesar salad
lime tahini salad dressing
gyro burger with greek salad
german yogurt dressing
pasta salad with smoked salmon and fava beans
roasted beet salad with orange dressing
green salad with posole and creamy cilantro lime vinaigrette

Cluster 7

simple and good sloppy joes
vegetarian swedish meatballs
radiatore sausage bake
beyond basic burgers 4 different burgers
hot pizza sub sandwich
garlic cheese sausage bites
quick taco dinner
skillet taco mac n cheese
low carb slaw
mediterranean pork and orzo

Cluster 8

soy glazed chicken thighs
bea s buffalo chicken wings white meat
miracle chicken skillet
chicken with white wine and olives
whole lemon roasted chicken
lime chicken burritos
honey glazed chicken breasts with rosemary and toasted almonds
chili plum stir fry chicken
down home chicken
penne with chicken and spinach and mushroom

Cluster 9

pomegranate cosmopolitans
masala chai tea
samosa
praline supreme latte
yellow bellied sap sucker
mike s peach puff martini
5 cup ambrosia
midori mai tai
dark n stormy
hungry girl s magical low calorie margarita

Cluster 10

mademoiselle grey cookies

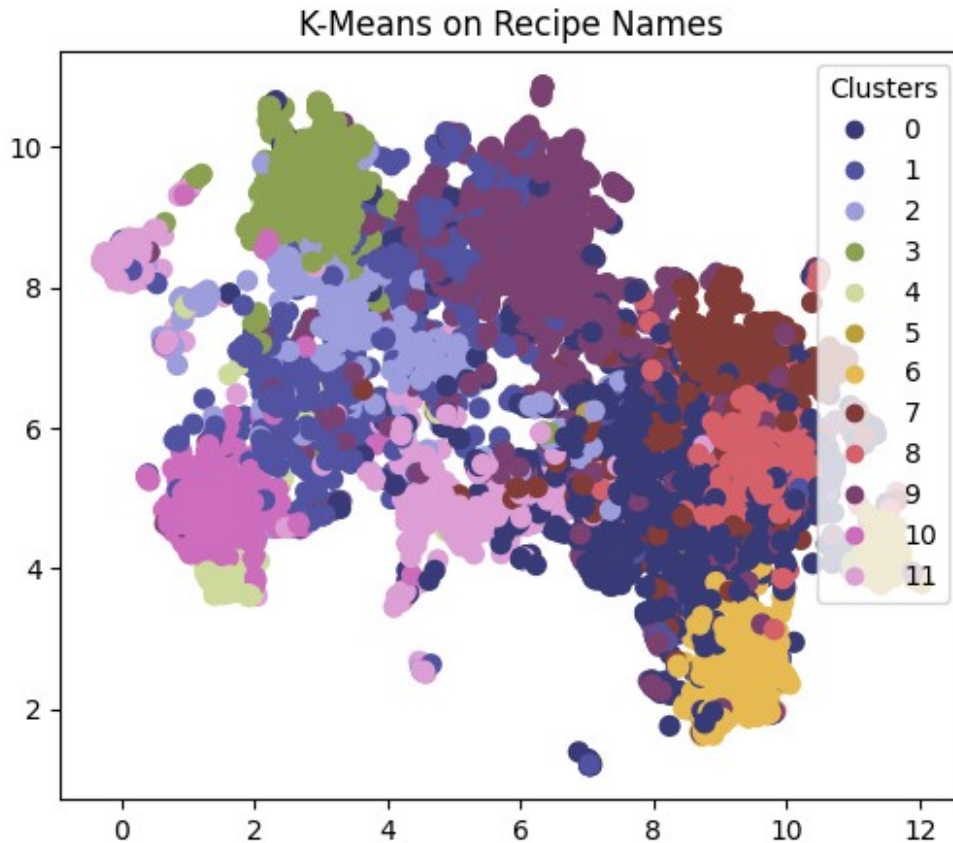
```
classic crunchy chocolate chip cookies
white lily chocolate dipped peanut butter cookies
oreo chunk cookies
light sugar cookies
large batch sugar cookies
fig cookies
evil peanut butter cookies
orange iced cranberry cookies
chewiest ever chocolate y chip ity cookies
```

Cluster 11

```
-----
fruit veggie muffins
italian shortbread with almonds and jam
savory bacon muffins
corn muffins by ina
rosemary goat cheese muffins
eggnog muffins with nutmeg streusel topping
low fat healthy zucchini bread
whole wheat cherry scones
gift of the magi bread
white chocolate iced cranberry bread
```

```
plot_umap_clusters(
    embeddings,
    kmeans_labels,
    raw_texts=recipes_df["name"].tolist(),
    title="K-Means on Recipe Names"
)
```

```
c:\Users\zhouz\miniforge3\envs\cpssc330-fresh\lib\site-packages\umap\
umap_.py:1952: UserWarning: n_jobs value 1 overridden to 1 by setting
random_state. Use no seed for parallelism.
    warn(
```

```
for k in [8, 10, 12, 15, 20]:  
    kmeans = KMeans(n_clusters=k, random_state=42)  
    labels = kmeans.fit_predict(embeddings)  
    print(f"k = {k}")  
    print_clusters(recipes_df, labels, n_recipes=5, random_state=42)
```

k = 8

Cluster 0

apricot crumble squares
spice and herb marinade
gluten free tropical rice
made from scratch broccoli cornbread
southwest style potato corn chowder

Cluster 1

isle of coconut
texas lizzies
spicy gingerman cocktail
sinful bars

truly the best key lime pie martini

Cluster 2

brunch egg burritos
red pepper biscuits
buttermilk blueberry pancakes
california white bread
texas roadhouse buttery dinner rolls with cinnamon butter

Cluster 3

heavenly white cake
german chocolate cake
fresh cran apple cake
butter rum cake from scratch
the best louisiana crunch cake ever

Cluster 4

nestle toll house mini morsel cookies
kairos chocolate chip cookies
dr cookie peanut butter squares
the barefoot contessa s ultimate ginger cookies
drop ginger cakes cookies

Cluster 5

creamed chicken with mushrooms pearl onions sherry
apple and onion chicken
aunt jo s crescent chicken rolls
chicken noodle bowl
takeout lemon chicken

Cluster 6

jan s broccoli salad
jen s taco salad
spicy strawberry spinach salad
chopped tandoori chicken salad
taco salad low carb

Cluster 7

beef fajita burgers
steak sandwiches with crispy onions
morrocan burgers with spicy mayo
dhingri gosht beef with bell peppers onions and mushrooms
l o v e wrap sandwich

k = 10

Cluster 0

pesto and potato tart
armenian pumpkin and lamb stew
cream cheese bacon spread
cheddary vegetable gratin
cheddar zucchini puff from better homes gardens

Cluster 1

lemon and blueberry yogurt parfait low fat
peanut butter meltaways
marshmallow fudge bars
butter mint shortbread
delicious marble bars

Cluster 2

spicy orange ricotta pancakes drizzled with rhubarb syrup
strawberry streusel cheesecake
tri chocolate mousse with raspberry sauce and chocolate filigree
grace s light fluffy cupcakes
pumpkin whoopie pies with cream cheese filling

Cluster 3

buckeye cake
peanut butter jelly snack cake
raspberry chocolate chip bundt cake
german bundt cake
weight watchers gob cake

Cluster 4

venetian cookies
kelly s amazing andes mint thumbprint cookies
barefoot contessa pb choc chunk cookies
iced lemon poppy seed cookies
drop sugar cookies

Cluster 5

pasta with no cook tomato mozzarella sauce
canadian killer kool aid recipe
poached salmon with sweet onions and balsamic vinaigrette
basil lemon garlic bread
greek roasted red pepper and feta dip

Cluster 6

easy veggie salad with asian dressing
black bean salad with sun dried tomatoes
tapenade caesar salad with prosciutto
chicken and peach salad
summertime potato salad

Cluster 7

1950 s meatloaf
peruvian quinoa stew vegan vegetarian
alvera s quick sauerbraten meatballs
old school meets new cheesy onion spicy bbq burger
victory s kicked up crown royal steak marinade

Cluster 8

cider braised chicken over smoked cheddar grits
honey lime chicken
old el paso baked spicy chicken tacos
mesquite grilled chicken with pineapple salsa
crispy oven baked chicken breasts

Cluster 9

dutch velvet
ginger limeade
adult cherry cola drink
goombay smash
hgs super skinny skins ww points 2

k = 12

Cluster 0

eliz s mushroom risotto
greet the sun breakfast pizzas
stuffed shells bolognese
creamy herb mashed potatoes
mast o musir persian yogurt dip with shallots

Cluster 1

butter chocolate bliss martini
peanut butter finger bars
kellogg s frosted flakes good times bars
chocolate mint brownies
goldy bear s honey gingersnaps

Cluster 2

berry cherry banana shortcake
chillin strawberry pie
drumstick cheesecake
lemon whoopie pies
white chocolate rhubarb swirled cheesecake

Cluster 3

calamondin cake
charlene s 7 up pound cake
corn bread cake
lemon yoghurt cake
coconut mocha cake

Cluster 4

linzie s oatmeal raisin delights
oatmeal cookies with white chocolate peanuts and caramel
kossmans oatmeal cookies
whole grain steel cut oatmeal pancakes
oatmeal applesauce muffins

Cluster 5

creamy mushroom roasted onion soup
spicy tomato beef soup
machine shed restaurant baked potato soup
potato leek and roasted garlic soup
rich beefy vegetable soup

Cluster 6

honey and garlic dressing 2
baby green salad
tapenade caesar salad with prosciutto
enlightened caesar salad
lime tahini salad dressing

Cluster 7

simple and good sloppy joes
vegetarian swedish meatballs
radiatore sausage bake
beyond basic burgers 4 different burgers
hot pizza sub sandwich

Cluster 8

soy glazed chicken thighs
bea s buffalo chicken wings white meat
miracle chicken skillet
chicken with white wine and olives
whole lemon roasted chicken

Cluster 9

pomegranate cosmopolitans
masala chai tea
samosa
praline supreme latte
yellow bellied sap sucker

Cluster 10

mademoiselle grey cookies
classic crunchy chocolate chip cookies
white lily chocolate dipped peanut butter cookies
oreo chunk cookies
light sugar cookies

Cluster 11

fruit veggie muffins
italian shortbread with almonds and jam
savory bacon muffins
corn muffins by ina
rosemary goat cheese muffins

k = 15

Cluster 0

barley pilaf with chickpeas and artichoke hearts
hot n hearty potatoes
potato skins filled with ham baked beans and cheese
belgium slice
jalapeno potatoes

Cluster 1

southwestern creamy ranch dip
hpnotiq margarita
ginger honey
cafe mandarin
lemon coconut colada

Cluster 2

cinnamon roll cream pie
spiced pumpkin cheesecake
baked egg boats
blueberry pineapple pina colada pie
turtle cheesecake des amis

Cluster 3

mini chocolate lava cakes
yellow cake
german chocolate cake with coconut pecan sauce
fig preserves cake
lil s banana cake with char s chocolate frosting

Cluster 4

blueberry oatmeal cookies
low cal low fat oatmeal carrot cookies
socks off oatmeal cookies no raisins
oatmeal craisins toffee chocolate cookies
oatmeal lace cookies with dried tart cherries

Cluster 5

basque lentil potato soup
italian chicken veggie soup paula deen
must love cheese chicken wild rice soup
cream of chicken soup
yucatan style chicken and vegetable soup

Cluster 6

chef salad dressing
jo s broccoli salad
mandie s chicken salad
apple pear and candied pecan salad
tunisian salad

Cluster 7

modified mas macho meatballs
perfect meatballs
the best ever beef tenderloin filet of beef
barb s sauteed swiss chard
oven meatballs and gravy

Cluster 8

chicken parmesan bundles
ww balsamic chicken with garlic couscous
flemish chicken stew with barley
calypso chicken
baked chicken chimichangas with green sauce

Cluster 9

blackout baked alaska
the west coast
marco polenta
orange kissed turnip custards
mock filet mignon

Cluster 10

iced sugar cookies cake like cookies soft icing
girl scout cookie from 1917
dks buttermilk biscuits
easy pumpkin cookies
healthy kashi cookies

Cluster 11

double corn and green chile muffins
tutti frutti muffins
asiago cheese muffins
tangerine muffins
mom s banana chocolate chip muffins

Cluster 12

mimi s cafe carrot raisin bread
banana bran scones cooking light
paleo pumpkin bread elana s pantry
sauerkraut rye bread sugar free
ferrycarrig traditional irish brown bread

Cluster 13

libbie s chocolate pudding
chocolate softies
coffee brownies
best ever brownies 6 ways
buttermilk bars

Cluster 14

wild rice and mushroom pilaf
jalapeno cornmeal pancakes
herbed cheese stuffed salmon
pizza in a pizza
aromatic noodles with lime peanut sauce

k = 20

Cluster 0

cheddar rice side dish
shrimp with spicy bloody mary sauce
pumpkin bacon and sundried tomato pasta bake
slow roast tomatoes
spinach and ricotta ravioli

Cluster 1

vanilla dreamsicle
dark and stormy
twisted snow pea toss
blizzard
hot macadamia spread

Cluster 2

hot fudge mocha pudding cake
streusel crumb coffee cake
peanut butter morsels cake
chocolate peanut butter gooey butter cake
cornstarch cake

Cluster 3

luscious lemon pound cake
banana cake made simple
pig picking cake
black forest cherry cake schwarzwlder kirchtorte
my favorite sour cream pound cake

Cluster 4

oat bran oatmeal cookies
brown sugar oatmeal pie
oatmeal bannock low fat
mina newton oatmeal raisin cookies
oatmeal toffee cookies

Cluster 5

mango pineapple smoothie
toffee almond sandies
oak alley s sweet mint iced tea
acapulco mocha iced latte
cinnamon and lemon tea

Cluster 6

spinach salad with orange vinaigrette
diva salad
not your momma s potato salad
blueberry spinach salad with chicken pecans and bleu cheese
scrumptious caesar salad dressing

Cluster 7

turkey cheeseburgers
chopped meat and beans
cube steaks and gravy
the original best ever meatloaf
manchego stuffed dates wrapped in bacon tapas

Cluster 8

baked chicken and mushrooms
creamy chicken cranberry pecan wild rice
avocado chicken melt
fresh pineapple and lychee chicken empress of china san fran
bea s buffalo chicken and blue cheese pizza

Cluster 9

mike s butterscotch martini
happily ever after cocktail
martini rossi rosso
godiva raspberry martini
bourbon stout yam waffles

Cluster 10

chocolate coffee toffee cookies
grannie s peanut butter cookies
orange glazed carrot cookies
key lime jewel cookies
brown sugar cookies

Cluster 11

kahlua and cream muffins
fruit veggie muffins
barefoot contessa s blueberry coffee cake muffins
cranberry grand marnier and almond muffins
ham and cheese buttermilk muffins

Cluster 12

cheesy herbed garlic bread

scottish bakers shortbreads 1959
hot melty oven baked ham swiss sandwiches
dr michael s cornbread abm
stuffed french toast ii

Cluster 13

protein powder brownies
blond toffee brownies
chocolate cherry fudge bundt
best ever blond brownies
peppermint brownie cake

Cluster 14

pastel de tres leches
giada s lasagna rolls
julie s turkey tetrazzini
la bomba
tostadas salsa verde

Cluster 15

juli neni s chicken soup
southwestern chicken barley soup
fennel avgolemono soup
quick easy mexican soup
potato and wild leek soup

Cluster 16

apple pie frosty protein smoothie
choc orange bread butter pudding
chocolate fried pies
shortcake with strawberry and raspberry
greek custard with phyllo pastry

Cluster 17

carrot bars
brigadeiros chocolate fudge truffles

```
special k bars
chocolate mint bars  a lighter version
peanut butter blocks
```

Cluster 18

```
-----
greek style stuffed bell peppers  light
kittencal s 60  whole wheat rolls
fish tortillas
spinach and artichoke dip  vegan
grilled salmon with orange glaze
```

Cluster 19

```
-----
favorite sweet potato and corn chowder
red garlic mashed potatoes
flemings potatoes
never fail scalloped potatoes
pork chops with potatoes and mushrooms
```

...

Ellipsis

Points:6

I applied DBSCAN with cosine distance and tried different values of eps and min_samples. With small eps values, most recipes were labeled as noise. After increasing eps, some clusters appeared but the results were still uneven. Overall, DBSCAN was harder to tune compared to KMeans.

```
from sklearn.cluster import DBSCAN
import numpy as np

for eps in [0.15, 0.2, 0.25, 0.3]:
    for min_samples in [3, 5, 10]:
        dbscan = DBSCAN(eps=eps, min_samples=min_samples,
metric="cosine")
```

```

labels = dbscan.fit_predict(embeddings)
n_clusters = len(set(labels)) - (1 if -1 in labels else 0)
n_noise = np.sum(labels == -1)
print(f"eps={eps}, min_samples={min_samples},
clusters={n_clusters}, noise={n_noise}")

eps=0.15, min_samples=3, clusters=136, noise=8259
eps=0.15, min_samples=5, clusters=30, noise=8687
eps=0.15, min_samples=10, clusters=4, noise=8924
eps=0.2, min_samples=3, clusters=215, noise=6958
eps=0.2, min_samples=5, clusters=68, noise=7760
eps=0.2, min_samples=10, clusters=12, noise=8497
eps=0.25, min_samples=3, clusters=164, noise=5130
eps=0.25, min_samples=5, clusters=61, noise=5968
eps=0.25, min_samples=10, clusters=19, noise=7059
eps=0.3, min_samples=3, clusters=85, noise=3201
eps=0.3, min_samples=5, clusters=25, noise=3757
eps=0.3, min_samples=10, clusters=11, noise=4820

dbscan = DBSCAN(eps=0.25, min_samples=5, metric="cosine")
dbscan_labels = dbscan.fit_predict(embeddings)

print_clusters(recipes_df, dbscan_labels, n_recipes=10, replace=True,
random_state=42)

```

Cluster -1

best ever zatarain s jambalaya
mushy peas
mini orange pochettes by melissa d arabian
mini bbq sausage tartlets
greek style marinated mushrooms
easy smoked turkey sausage cassoulet
orange pumpkin scone cookie cut outs
bacon and red potato bake
roasted red potatoes with shallots and fresh herbs
kahlua cognac

Cluster 0

buttermilk ranch dressing
honey and garlic dressing 2
raspberry vinaigrette salad dressing
tropical lime chicken
basil garlic vinaigrette
another garlic chicken
caterer s basic french salad dressing
red leaf salad

taylor s landing spinach salad with honey mustard dressing
easy honey french dressing

Cluster 1

new york deli style cheesecake
new york cheesecake
cheesecake supreme
new york deli style cheesecake
cheater s new york style cheesecake
philadelphia oreo cheesecake
cheesecake supreme
cheesecake supreme
new york deli style cheesecake
new york style crustless cheesecake

Cluster 2

potato leek and roasted garlic soup
lentil and barley soup
tomato and red lentil soup
kelly s tlc chicken noodle soup
mushroom and pea soup
chicken and lime soup
potato leek and roasted garlic soup
yucatan style chicken and vegetable soup
ground beef and vegetable soup easy and delicious
mexican rice and bean soup

Cluster 3

favorite homemade buttermilk biscuits
cinnamon bun cake
banana coconut tea bread
dads cookies
spicy molasses crinkle cookies
frosted brown sugar pound cake
carrot cake supreme
giant oat cookies
chocolate caramel oatmeal bars
vanilla sour cream cupcakes small batch

Cluster 4

naughtiest most delicious spinach artichoke dip
spinach and artichoke dip vegan
creamy artichoke spinach crab dip
spinach and artichoke dip vegan
spinach and artichoke dip vegan
artichoke spinach dip
creamy artichoke spinach crab dip
creamy artichoke spinach crab dip
creamy artichoke spinach crab dip
spinach and artichoke dip vegan

Cluster 5

spiced peach cocktail
key lime martini
caribbean martini
pomegrante martini
champagne martini
peaches n cream martini
blue margarita
a berry lime martini
butterscotch truffle martini
butterscotch truffle martini

Cluster 6

easy breakfast casserole ole
low fat country style breakfast casserole
bo s breakfast casserole
low fat country style breakfast casserole
low fat country style breakfast casserole
amazing breakfast casserole
bo s breakfast casserole
bo s breakfast casserole
bo s breakfast casserole
low fat country style breakfast casserole

Cluster 7

cheesy garlic bread
ch easy golden garlic bread
herb quick bread

frank s cheesy garlic bread
cheesy herbed garlic bread
cheesy garlic bread
weight watchers garlic herb bread
garlic herb bread
cheesy garlic bread
herb quick bread

Cluster 8

apple pecan bread pudding
apple butter pecan cake
pecan chip tube cake
apple butter pecan cake
apple butter pecan cake
raspberry brandy pecan cake
pecan chip tube cake
pecan chip tube cake
pecan chip tube cake
apple butter pecan cake

Cluster 9

spicy apricot baby food cake
apricot and almond snack cake
apricot gooey cake
apricot and almond snack cake
apricot and almond snack cake
apricot nectar cake
apricot gooey cake
apricot gooey cake
apricot gooey cake
apricot and almond snack cake

Cluster 10

the original best ever meatloaf
cheeseburger meatloaf with sauce
german meatloaf
lower fat totally awesome meatloaf
best ever meatloaf
the original best ever meatloaf
macaroni and cheese meatloaf
turkey meatloaf

ranch meatloaf
ranch meatloaf

Cluster 11

baked chicken and dumplings
chicken and cornmeal dumplings
cooked chicken for dumplings
chicken and dumplings for 2
baked chicken and dumplings
awesome chicken and dumplings
baked chicken and dumplings
cooked chicken for dumplings
chicken and dumplings for 2
chicken and cornmeal dumplings

Cluster 12

mini chicken meatballs
swedish meatballs with secret ingredient
delicious italian meatballs
best meatballs
my favorite spaghetti and meatballs
mini chicken meatballs
swedish meatballs awesome
best meatballs
best meatballs
italian spaghetti with meatballs

Cluster 13

celestial french toast
capn crunch french toast
waffled french toast
strawberry and cream cheese stuffed french toast
custardy french toast
stuffed french toast casserole
celestial french toast
strawberry mascarpone french toast
mixed berry french toast bake
strawberry and cream cheese stuffed french toast

Cluster 14

beer marinade for steak
red wine marinade
red wine herb marinated beef steak
red wine marinade
red wine marinade
beef in red wine marinade
red wine herb marinated beef steak
red wine herb marinated beef steak
red wine herb marinated beef steak
red wine marinade

Cluster 15

easy farmhouse multi grain bread for bread machine
dark rye pumpnickel bread for the bread machine
cornell bread bread machine
beer rye bread
swedish rye bread bread machine
sweet rye bread
cornell bread bread machine
sourdough beer bread bread machine
easy farmhouse multi grain bread for bread machine
milk bread bread machine

Cluster 16

ihop style pancakes
down home pancakes
best ever homemade pancakes
ihop style pancakes
the best pancakes in the world
best ever homemade pancakes
best ever homemade pancakes
ihop style pancakes
pancakes best ever bisquick
the best pancakes in the world

Cluster 17

spelt biscuits
hope s bisquick biscuits
mimi s biscuits

my husband s biscuits
spelt biscuits
mimi s biscuits
mimi s biscuits
orange tea biscuits
my husband s biscuits
ray gregg s batch biscuits southern style

Cluster 18

herbed chickpea salad
white bean and chickpea salad
chickpea salad with roasted cumin
herbed chickpea salad
key west black bean salad
black bean and corn salad ww core
chickpea salad with roasted cumin
chickpea salad with roasted cumin
herbed chickpea salad
quick chickpeas salad

Cluster 19

new years fruity smoothie
black and blue cranberry smoothie
smooth mango lemonade
nectarine pineapple smoothie
new years fruity smoothie
smooth mango lemonade
smooth mango lemonade
strawberry yogurt smoothie
nectarine pineapple smoothie
green tea smoothie

Cluster 20

country chicken
chicken fried steak and gravy
zesty pan fried chicken
breaded fried cube steak and milk gravy
ultra low fat chicken fried chicken with cream gravy
country fried chicken
zesty pan fried chicken
cube steaks and gravy

country chicken
country fried cube steak and gravy

Cluster 21

swiss chicken casserole
buffalo chicken casserole
cheesy chicken noodle casserole
creamy swiss chicken
creamy noodle casserole
chicken breast casserole
swiss chicken casserole
buffalo style chicken pizza
creamy swiss chicken
creamy swiss chicken

Cluster 22

buttermilk spice cake
chai buttermilk spice cake
buttermilk spice cake
buttermilk spice cake
chai buttermilk spice cake
chai buttermilk spice cake
buttermilk spice cake
buttermilk maple spice cake
buttermilk spice cake
buttermilk spice cake

Cluster 23

taco salad low carb
jen s taco salad
california taco salad
taco salad low carb
chicken taco slaw salad
california taco salad
california taco salad
taco salad low carb
midwestern taco salad
chicken taco slaw salad

Cluster 24

herbal iced tea
lemon lime iced tea
electric iced tea
pineapple mint iced tea
herbal iced tea
cherry mint iced tea
mcdonald s vanilla iced coffee
herbal iced tea
electric iced tea
pineapple mint iced tea

Cluster 25

spicy moroccan chicken skewers
chicken meatballs and spicy peanut sauce
chicken skewers with spicy peanut sauce
chicken meatballs and spicy peanut sauce
chicken meatballs and spicy peanut sauce
sesame chicken skewers
chicken skewers with spicy peanut sauce
chicken skewers with spicy peanut sauce
chicken skewers with spicy peanut sauce
chicken meatballs and spicy peanut sauce

Cluster 26

macaroons
coconut cranberry macaroon
coconut macaroons or macaroon kisses
coconut cranberry macaroon
coconut cranberry macaroon
chocolate coconut macaroons
coconut macaroons or macaroon kisses
coconut macaroons or macaroon kisses
coconut macaroons or macaroon kisses
coconut cranberry macaroon

Cluster 27

chopped beef stroganoff
simple beef stroganoff
low carb beef stroganoff

chopped beef stroganoff
quick easy beef stroganoff
annie s fabulous beef stroganoff
low carb beef stroganoff
low carb beef stroganoff
chopped beef stroganoff
beef stroganoff meatballs

Cluster 28

mixed greens with goat cheese and candied almonds
cold tomato cheese salad
mixed greens with pecans goat cheese and dried cranberries
cold spaghetti salad
mixed greens with goat cheese and candied almonds
tomato salad with goat cheese
mixed greens with goat cheese and candied almonds
mixed greens with pecans goat cheese and dried cranberries
cold spaghetti salad
cold tomato cheese salad

Cluster 29

sarah s tamale pie with cornbread crust
super tamale pie
tasty tamale pie casserole
super tamale pie
super tamale pie
easy tamale casserole
tasty tamale pie casserole
tasty tamale pie casserole
tasty tamale pie casserole
super tamale pie

Cluster 30

creamy chipotle cabbage slaw
red cabbage coleslaw rotkohlsalat
creamy no fat coleslaw
red cabbage coleslaw rotkohlsalat
red cabbage coleslaw rotkohlsalat
differently delicious coleslaw
creamy no fat coleslaw
creamy no fat coleslaw

creamy no fat coleslaw
red cabbage coleslaw rotkohlsalat

Cluster 31

mrs galumpki s stuffed cabbage rolls
stuffed cabbage rolls kaldolmar
unrolled cabbage rolls
mrs galumpki s stuffed cabbage rolls
star s cabbage rolls
unrolled cabbage rolls
unrolled cabbage rolls
mrs galumpki s stuffed cabbage rolls
stuffed cabbage rolls with sweet and sour sauce
star s cabbage rolls

Cluster 32

deliciously soft gingersnaps
good old gingersnaps
deliciously soft gingersnaps
deliciously soft gingersnaps
good old gingersnaps
good old gingersnaps
deliciously soft gingersnaps
favorite gingersnaps
deliciously soft gingersnaps
deliciously soft gingersnaps

Cluster 33

stuffed red bell pepper
easy stuffed green peppers
southwest stuffed bell peppers
ww hearty stuffed bell peppers
barbecue style stuffed green peppers
southwest stuffed bell peppers
jasmine rice stuffed bell peppers
stuffed red bell pepper
sarasota s chorizo and black bean stuffed cubanelle peppers
inside out stuffed bell peppers

Cluster 34

roasted garlic and herb mash potatoes
herb roasted sweet potatoes
creamy herb mashed potatoes
oven roasted red potatoes with herbs
roasted potatoes with whole garlic and rosemary
rosemary roasted red potatoes
creamy herb mashed potatoes
make ahead mashed potatoes
roasted garlic and herb mash potatoes
roasted potatoes with garlic crust

Cluster 35

country style greek salad
roasted garlic and sun dried tomato dressing for greek salad
greek grilled chicken salad
roasted garlic and sun dried tomato dressing for greek salad
roasted garlic and sun dried tomato dressing for greek salad
greek grilled pita salad
greek grilled chicken salad
greek grilled chicken salad
greek grilled chicken salad
roasted garlic and sun dried tomato dressing for greek salad

Cluster 36

white bean and apple chili
buckaroo chicken green chilies and rice rsc
green bay chili
white bean and apple chili
white bean and apple chili
buckaroo chicken green chilies and rice rsc
green bay chili
green bay chili
white bean and apple chili
green chili with white beans aka white bean chicken chili

Cluster 37

lager and lemon limeade
vanilla lemonade
hot lemonade
vanilla lemonade

vanilla lemonade
hard strawberry lemonade
hot lemonade
hot lemonade
hot lemonade
vanilla lemonade

Cluster 38

quick unbaked cheesecake
easy sopapilla cheesecake
impossibly easy cheesecake
easy sopapilla cheesecake
easy sopapilla cheesecake
impossibly easy toffee bar cheesecake
impossibly easy cheesecake
impossibly easy cheesecake
impossibly easy cheesecake
easy sopapilla cheesecake

Cluster 39

caesar crusted chicken
sicilian chicken
italian boneless chicken
sicilian chicken
sicilian chicken
italian crusted chicken
italian boneless chicken
italian boneless chicken
italian boneless chicken
sicilian chicken

Cluster 40

marshmallow fudge cookies
minty cocoa fudge sandwich cookies
chocolate marshmallow meltaway cookies
marshmallow fudge cookies
marshmallow fudge cookies
minty cocoa fudge sandwich cookies
chocolate marshmallow meltaway cookies
chocolate marshmallow meltaway cookies
marshmallow fudge cookies

fudge cookies

Cluster 41

zingy meat loaf
triple threat meat loaf
good easy meat loaf
zingy meat loaf
ham and mushroom meat loaf
salsa meat loaf
good easy meat loaf
good easy meat loaf
zingy meat loaf
favorite meat loaf vegan

Cluster 42

quick taco dinner
mexican dinner aka taco pie
mexican burrito pie
mexican dinner aka taco pie
mexican dinner aka taco pie
southwest taco pie
mexican burrito pie
mexican burrito pie
mexican burrito pie
mexican dinner aka taco pie

Cluster 43

no bake cheesecake
no bake toblerone cheesecake
no bake chocolate peanut butter cheesecake
no bake toblerone cheesecake
no bake toblerone cheesecake
rich no bake cheesecake
no bake chocolate peanut butter cheesecake
no bake chocolate peanut butter cheesecake
no bake chocolate peanut butter cheesecake
no bake toblerone cheesecake

Cluster 44

country potato pie
old fashioned sweet potato pie
old fashioned cinnamon candy apple pie
country potato pie
country potato pie
old fashioned sweet potato pie
old fashioned cinnamon candy apple pie
old fashioned cinnamon candy apple pie
country potato pie
pass that sweet potato pie

Cluster 45

citrus spiced coffee
spiced citrus coffee
orange spiced coffee
spiced citrus coffee
spiced citrus coffee
pumpkin spice coffee
orange spiced coffee
orange spiced coffee
orange spiced coffee
spiced citrus coffee

Cluster 46

grilled oriental chicken
maple and orange glazed chicken
orange molasses glazed chicken
maple and orange glazed chicken
maple and orange glazed chicken
oriental glazed chicken
orange molasses glazed chicken
orange molasses glazed chicken
orange molasses glazed chicken
maple and orange glazed chicken

Cluster 47

this cake almost tastes like a pina colada
pina colada poke cake
amazingly delicious rum pina colada cake
pina colada poke cake

pina colada poke cake
pina colada trifle
amazingly delicious rum pina colada cake
amazingly delicious rum pina colada cake
amazingly delicious rum pina colada cake
pina colada poke cake

Cluster 48

pink lemonade alcoholic
pink lemonade thins
pink lemonade spritzer
pink lemonade thins
pink lemonade thins
pink lemonade spritzer
pink lemonade spritzer
pink lemonade spritzer
pink lemonade spritzer
pink lemonade thins

Cluster 49

pink floyd cocktail
pink vodka cocktail
pink prozac cocktail
pink vodka cocktail
pink vodka cocktail
pink sauterne cocktail
pink prozac cocktail
pink prozac cocktail
pink prozac cocktail
pink vodka cocktail

Cluster 50

pork medallions in apple mustard sauce
pork tenderloin medallions with fresh figs
florentine pork medallions
pork tenderloin medallions with fresh figs
pork tenderloin medallions with fresh figs
pork medallions with wine and sage
florentine pork medallions
florentine pork medallions
florentine pork medallions

pork tenderloin medallions with fresh figs

Cluster 51

mediterranean shrimp n pasta
quick shrimp and pasta
nif s baked pasta with shrimp and chicken
quick shrimp and pasta
quick shrimp and pasta
another quick shrimp and chorizo paella
nif s baked pasta with shrimp and chicken
nif s baked pasta with shrimp and chicken
nif s baked pasta with shrimp and chicken
quick shrimp and pasta

Cluster 52

delicious and simple baked salmon with fancy sauce
salmon cakes with chili mayonnaise sauce
salmon burgers patties with a herb sauce
delicious and simple baked salmon with fancy sauce
peppercorn crusted salmon with white wine butter sauce
salmon with grainy mustard and herb sauce
salmon burgers patties with a herb sauce
salmon burgers patties with a herb sauce
delicious and simple baked salmon with fancy sauce
steamed salmon with chili and basil sauce

Cluster 53

easy scalloped potatoes and ham
scalloped potatoes etc
sharon s tasty scalloped potatoes
scalloped potatoes etc
scalloped potatoes etc
absolutely delightful scalloped potatoes
sharon s tasty scalloped potatoes
sharon s tasty scalloped potatoes
sharon s tasty scalloped potatoes
scalloped potatoes etc

Cluster 54

shepherd s vegetable pie
down home shepherd s pie
shepherdess pie
down home shepherd s pie
down home shepherd s pie
skillet shepherd s pie
shepherdess pie
shepherdess pie
shepherdess pie
down home shepherd s pie

Cluster 55

turkey velvet soup
turkey neck soup
simple turkey soup
turkey neck soup
turkey neck soup
tex mex turkey soup
simple turkey soup
simple turkey soup
simple turkey soup
turkey neck soup

Cluster 56

southwestern creamy ranch dip
simple spicy corn dip
ranch style dip
simple spicy corn dip
simple spicy corn dip
chili s avocado ranch dipping sauce
ranch style dip
ranch style dip
ranch style dip
simple spicy corn dip

Cluster 57

spicy noodles with ginger garlic shrimp and wasabi sauce
aromatic noodles with lime peanut sauce
spicy sesame noodles with chopped peanuts thai basil
aromatic noodles with lime peanut sauce

aromatic noodles with lime peanut sauce
thai pork chops and sesame noodles
spicy sesame noodles with chopped peanuts thai basil
spicy sesame noodles with chopped peanuts thai basil
spicy sesame noodles with chopped peanuts thai basil
aromatic noodles with lime peanut sauce

Cluster 58

tomato onion soup
tomato soup quick and easy
tomato onion soup
tomato onion soup
tomato soup quick and easy
tomato soup quick and easy
tomato onion soup
sherried tomato soup
tomato onion soup
tomato onion soup

Cluster 59

cheesy pork burgers
thai turkey burgers
ww turkey and cheddar burgers 7 points
thai turkey burgers
thai turkey burgers
turkey cheddar burgers
ww turkey and cheddar burgers 7 points
ww turkey and cheddar burgers 7 points
ww turkey and cheddar burgers 7 points
thai turkey burgers

Cluster 60

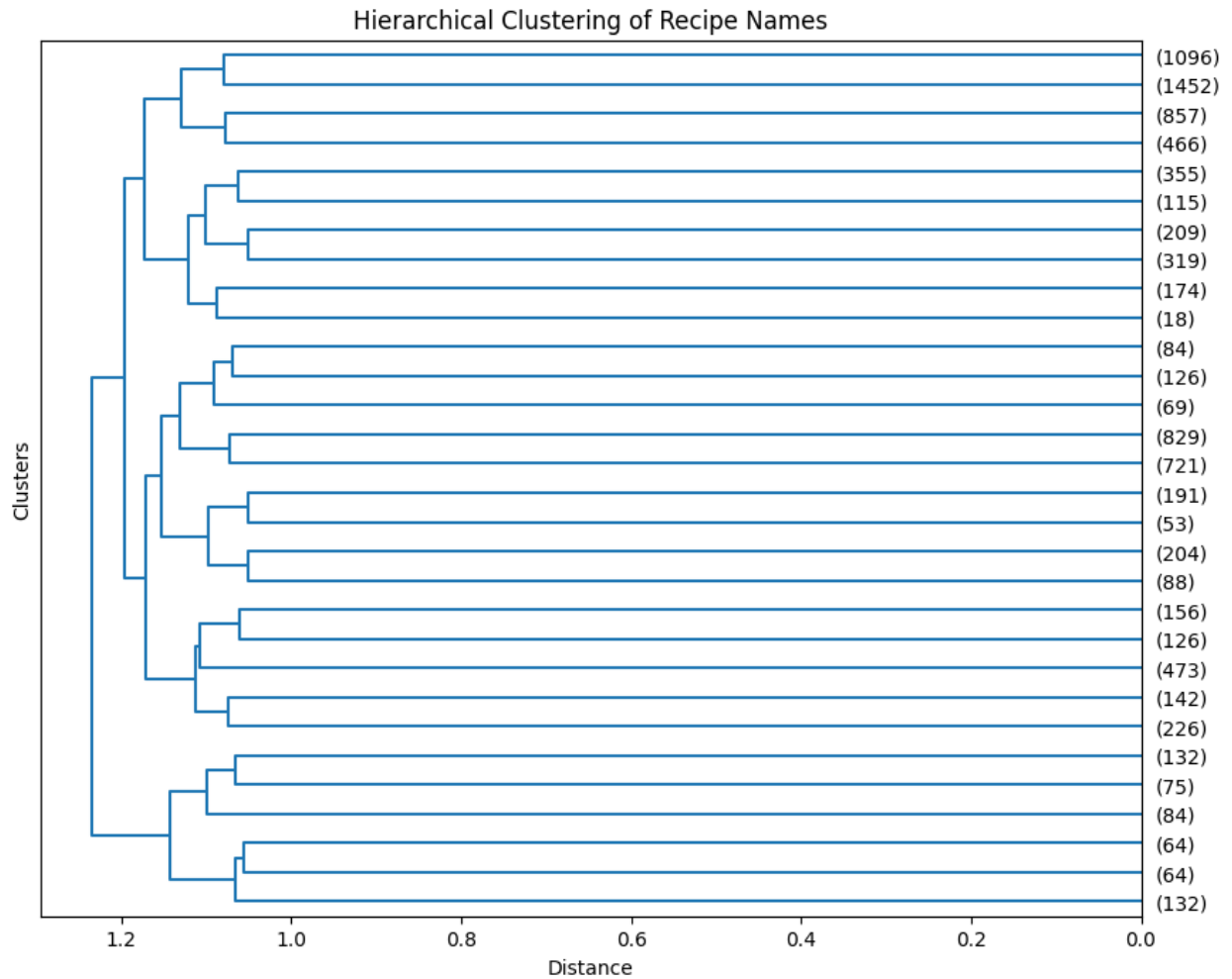
buffalo chicken twice baked potatoes
cupcake princess twice baked potatoes
herbed twice baked potatoes
cupcake princess twice baked potatoes
cupcake princess twice baked potatoes
leek and ricotta twice baked potatoes
herbed twice baked potatoes
herbed twice baked potatoes
herbed twice baked potatoes

cupcake princess twice baked potatoes

Points:6

Type your answer here, replacing this text.

```
from scipy.cluster.hierarchy import linkage
Z = linkage(embeddings, method="complete", metric="cosine")
from scipy.cluster.hierarchy import dendrogram
import matplotlib.pyplot as plt
plt.figure(figsize=(10,8))
dendrogram(
    Z,
    truncate_mode="lastp",
    p=30,
    orientation="left"
)
plt.title("Hierarchical Clustering of Recipe Names")
plt.xlabel("Distance")
plt.ylabel("Clusters")
plt.show()
```



```
from scipy.cluster.hierarchy import fcluster
hier_labels = fcluster(Z, t=12, criterion="maxclust")
print_clusters(recipes_df, hier_labels, n_recipes=5, random_state=42)
```

Cluster 1

```
-----
matcha green tea cookies
flaming angry dragon
fat santa s
new orleans hurricane
old fashioned
```

Cluster 2

```
-----
grapefruit cosmo
kato s chocolate hazelnut delight torte
```

hazelnut torte
lasagna with artichokes no cook noodles
pastel de tres leches

Cluster 3

simple spicy corn dip
tropical romaine salad with pineapple vinaigrette
mother s bavarian pot roast
chocolate irish cream cheesecake
julia child s master recipe for beef bourguignon in red wine

Cluster 4

van nuys aka dirty margarita
hard strawberry lemonade
margarita on the rocks
purple margarita
pockets of lemon cake

Cluster 5

lion s head meatballs
apple strips
strawberry cheese ball
white cheddar apple and bacon cheesecake
homesteader s fireweed honey

Cluster 6

quinoa pudding with apricots and pistachios
red snapper adult drink
lactose free indian curry
fusilli with mushrooms and chard
pink prozac cocktail

Cluster 7

greek tomatoes ww
gnocchi in rosa sauce
lemon pepper chicken with lemon wine sauce

parmesan chicken cutlets and popcorn chicken for the kids
mini apple dumplings

Cluster 8

chubby hubby meatloaf
nutty rocky road bars
bamieh okra and lamb middle east palestine
turkey meatloaf
pink lady squares

Cluster 9

cafe vanilla or cafe vanilla caramel
coffee caramel creme brulee
french vanilla frozen coffee drink
spiced citrus coffee
my version of caramel salted cocoa

Cluster 10

quinoa chicken fried rice
anna s granola cookies
bisquick butterscotch brownies
chicken floating in curried coconut soup
chocolate coconut crispies

Cluster 11

orange blossom zucchini
spiced garden salad dressing
mocha caramel fondue
greek salad with shredded chicken
make ahead spinach and boston lettuce salad

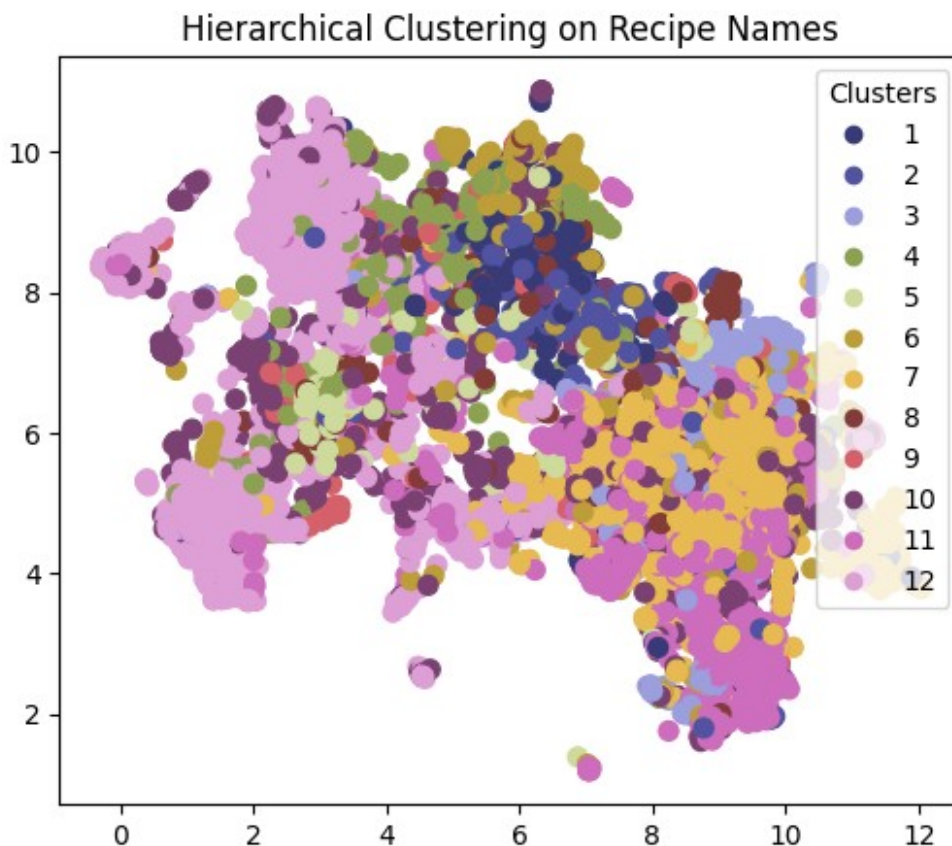
Cluster 12

potato chip chocolate chip cookies
oreo cookie cheesecake with chocolate glaze
black bottom banana cream pie light version
quick and favorite cake

```
strawberry disaster cake
```

```
plot_umap_clusters(  
    embeddings,  
    hier_labels,  
    raw_texts=recipes_df["name"].tolist(),  
    title="Hierarchical Clustering on Recipe Names"  
)
```

```
c:\Users\zhouz\miniforge3\envs\cpssc330-fresh\lib\site-packages\umap\  
umap_.py:1952: UserWarning: n_jobs value 1 overridden to 1 by setting  
random_state. Use no seed for parallelism.  
warn()
```



Points:6

Some clusters look like similar recipe types. For example dessert recipes like brownies or bars are often grouped together, while other clusters have pasta, rice or vegetable dishes.

There are some common themes across methods, like sweet foods vs main dishes. But the clusters are not exactly same. K-means gives more balanced groups, DBSCAN has many noise points, and hierarchical clustering shows structure but still mix some recipes.

Overall clustering works ok but recipe names are short so sometimes the grouping is not very clear.

Before submitting your assignment, please make sure you have followed all the instructions in the Submission Instructions section at the top.

Here is a quick checklist before submitting:

- ☐ Restart kernel, clear outputs, and run all cells from top to bottom.
- ☐ `.ipynb` file runs without errors and contains all outputs.
- ☐ Only `.ipynb` and required output files are uploaded (no extra files).
- ☐ Execution numbers start at **1** and are in order.
- ☐ If `.ipynb` is too large and doesn't render on Gradescope, also upload a PDF/HTML version.
- ☐ Reviewed the [CPSC 330 homework instructions](#).